

$$\begin{aligned} & \frac{1}{16\pi^2} \left(\frac{1}{24} s_\gamma c_\gamma \frac{1}{m_b^2} \left(3 y_e^{i1i2} y_u^{pi4} \left(\overline{y_d^{pr}} y_d^{i3r} \left(1 + 2 c_\gamma^2 + s_\gamma^2 \right) + 3 \overline{y_u^{pr}} y_u^{i3r} \left(1 + c_\gamma^2 \right) \right) + \right. \right. \\ & \quad y_u^{i3i4} \left(3 \overline{y_e^{pr}} \left(3 y_e^{pi2} y_e^{i1r} \left(1 + s_\gamma^2 \right) + 4 s_\gamma^2 y_e^{pr} y_e^{i1i2} \right) + \right. \\ & \quad \left. \left. y_e^{i1i2} \left(19 g_1^2 - 9 g_2^2 + 48 g_3^2 + 36 c_\gamma^2 \overline{y_u^{pr}} y_u^{pr} \right) \right) \right) + \\ & \frac{1}{2} \sum_p g_1^2 \frac{1}{m_b^4} y_e^{i1i2} y_u^{i3i4} \left(s_{2\gamma} c_\gamma^2 + s_\gamma c_{2\gamma} c_\gamma - s_{2\gamma} s_\gamma^2 \right) LF_{1,0}[m_d^p] + \\ & \frac{3}{2} \frac{1}{m_b^4} \overline{y_d^{pr}} y_d^{pr} y_e^{i1i2} y_u^{i3i4} \left(-s_{2\gamma} c_\gamma^2 + s_{2\gamma} s_\gamma^2 + 2 c_\gamma s_\gamma^3 \right) LF_{1,0}[m_d^r] + \\ & \frac{1}{2} \sum_p g_1^2 \frac{1}{m_b^4} y_e^{i1i2} y_u^{i3i4} \left(s_{2\gamma} c_\gamma^2 + s_\gamma c_{2\gamma} c_\gamma - s_{2\gamma} s_\gamma^2 \right) LF_{1,0}[m_e^p] + \\ & \frac{1}{2} \frac{1}{m_b^4} \overline{y_e^{pr}} y_e^{pr} y_e^{i1i2} y_u^{i3i4} \left(-s_{2\gamma} c_\gamma^2 + s_{2\gamma} s_\gamma^2 + 2 c_\gamma s_\gamma^3 \right) LF_{1,0}[m_e^r] + \frac{1}{2} \frac{1}{m_b^4} y_e^{i1i2} y_u^{i3i4} \\ & \quad \left(\overline{y_e^{pr}} y_e^{pr} \left(-s_{2\gamma} c_\gamma^2 + s_{2\gamma} s_\gamma^2 + 2 c_\gamma s_\gamma^3 \right) - \sum_p g_1^2 \left(s_{2\gamma} c_\gamma^2 + s_\gamma c_{2\gamma} c_\gamma - s_{2\gamma} s_\gamma^2 \right) \right) LF_{1,0}[m_l^p] + \\ & \frac{1}{2} \frac{1}{m_b^4} y_e^{i1i2} y_u^{i3i4} \left(3 \overline{y_d^{pr}} y_d^{pr} \left(-s_{2\gamma} c_\gamma^2 + s_{2\gamma} s_\gamma^2 + 2 c_\gamma s_\gamma^3 \right) + \right. \\ & \quad \left. 3 \overline{y_u^{pr}} y_u^{pr} \left(s_{2\gamma} c_\gamma^2 + 2 s_\gamma c_\gamma^3 - s_{2\gamma} s_\gamma^2 \right) + \sum_p g_1^2 \left(s_{2\gamma} c_\gamma^2 + s_\gamma c_{2\gamma} c_\gamma - s_{2\gamma} s_\gamma^2 \right) \right) LF_{1,0}[m_q^p] - \\ & \sum_p g_1^2 \frac{1}{m_b^4} y_e^{i1i2} y_u^{i3i4} \left(s_{2\gamma} c_\gamma^2 + s_\gamma c_{2\gamma} c_\gamma - s_{2\gamma} s_\gamma^2 \right) LF_{1,0}[m_u^p] + \\ & \frac{3}{2} \frac{1}{m_b^4} y_e^{i1i2} \overline{y_u^{pr}} y_u^{pr} y_u^{i3i4} \left(s_{2\gamma} c_\gamma^2 + 2 s_\gamma c_\gamma^3 - s_{2\gamma} s_\gamma^2 \right) LF_{1,0}[m_u^r] + \frac{1}{8} \frac{1}{m_b^4} y_e^{i1i2} y_u^{i3i4} \\ & \quad \left(3 s_{4\gamma} c_\gamma^2 \left(g_1^2 + g_2^2 \right) + 2 s_\gamma c_\gamma \left(g_1^2 \left(-1 + 3 c_{2\gamma}^2 \right) + 3 g_2^2 \left(-1 + c_{2\gamma}^2 \right) \right) - 3 s_{4\gamma} s_\gamma^2 \left(g_1^2 + g_2^2 \right) \right) \\ & \quad LF_{1,0}[m_\boxplus] + \frac{1}{4} s_\gamma c_\gamma \frac{1}{m_b^2} \left(3 y_e^{i1i2} y_u^{pi4} \left(s_\gamma^2 \overline{y_d^{pr}} y_d^{i3r} - c_\gamma^2 \overline{y_u^{pr}} y_u^{i3r} \right) + \right. \\ & \quad \left. y_u^{i3i4} \left(-3 s_\gamma^2 \overline{y_e^{pr}} y_e^{pi2} y_e^{i1r} + 2 y_e^{i1i2} \left(g_1^2 + 3 g_2^2 \right) \right) \right) LF_{1,1}[m_\boxplus] + \\ & \frac{1}{4} s_\gamma c_\gamma y_e^{i1i2} \left(2 c_\gamma^2 \overline{y_d^{pr}} y_d^{i3r} y_u^{pi4} - y_u^{i3i4} \left(g_1^2 + 3 g_2^2 \right) \right) LF_{1,2}[m_\boxplus] - \\ & s_\gamma c_\gamma g_1^2 \frac{1}{m_b^2} y_e^{i1i2} y_u^{i3i4} LF_{1,1,0}[m_1, m_e^{i2}] + \frac{1}{2} s_\gamma c_\gamma g_1^2 \frac{1}{m_b^2} y_e^{i1i2} y_u^{i3i4} LF_{2,1,-1}[m_1, m_e^{i2}] - \\ & \frac{1}{4} s_\gamma c_\gamma g_1^2 \frac{1}{m_b^2} y_e^{i1i2} y_u^{i3i4} LF_{1,1,0}[m_1, m_l^{i1}] + \frac{1}{8} s_\gamma c_\gamma g_1^2 \frac{1}{m_b^2} y_e^{i1i2} y_u^{i3i4} LF_{2,1,-1}[m_1, m_l^{i1}] - \\ & \frac{1}{36} s_\gamma c_\gamma g_1^2 \frac{1}{m_b^2} y_e^{i1i2} y_u^{i3i4} LF_{1,1,0}[m_1, m_q^{i3}] + \\ & \frac{1}{72} s_\gamma c_\gamma g_1^2 \frac{1}{m_b^2} y_e^{i1i2} y_u^{i3i4} LF_{2,1,-1}[m_1, m_q^{i3}] - \frac{4}{9} s_\gamma c_\gamma g_1^2 \frac{1}{m_b^2} y_e^{i1i2} y_u^{i3i4} LF_{1,1,0}[m_1, m_u^{i4}] + \\ & \frac{2}{9} s_\gamma c_\gamma g_1^2 \frac{1}{m_b^2} y_e^{i1i2} y_u^{i3i4} LF_{2,1,-1}[m_1, m_u^{i4}] - s_\gamma c_\gamma g_1^2 \frac{1}{m_b^4} y_e^{i1i2} y_u^{i3i4} LF_{1,1,-1}[m_1, \tilde{\mu}] - \\ & m_1 \tilde{\mu} g_1^2 \frac{1}{m_b^4} y_e^{i1i2} y_u^{i3i4} \left(c_\gamma^4 - 4 s_\gamma^2 c_\gamma^2 + s_\gamma^4 \right) LF_{1,1,0}[m_1, \tilde{\mu}] - \\ & \frac{3}{4} s_\gamma c_\gamma g_2^2 \frac{1}{m_b^2} y_e^{i1i2} y_u^{i3i4} LF_{1,1,0}[m_2, m_l^{i1}] + \\ & \frac{3}{8} s_\gamma c_\gamma g_2^2 \frac{1}{m_b^2} y_e^{i1i2} y_u^{i3i4} LF_{2,1,-1}[m_2, m_l^{i1}] - \frac{3}{4} s_\gamma c_\gamma g_2^2 \frac{1}{m_b^2} y_e^{i1i2} y_u^{i3i4} LF_{1,1,0}[m_2, m_q^{i3}] + \\ & \frac{3}{8} s_\gamma c_\gamma g_2^2 \frac{1}{m_b^2} y_e^{i1i2} y_u^{i3i4} LF_{2,1,-1}[m_2, m_q^{i3}] - 3 s_\gamma c_\gamma g_2^2 \frac{1}{m_b^4} y_e^{i1i2} y_u^{i3i4} LF_{1,1,-1}[m_2, \tilde{\mu}] - \\ & 3 m_2 \tilde{\mu} g_2^2 \frac{1}{m_b^4} y_e^{i1i2} y_u^{i3i4} \left(c_\gamma^4 - 4 s_\gamma^2 c_\gamma^2 + s_\gamma^4 \right) LF_{1,1,0}[m_2, \tilde{\mu}] - \\ & \frac{4}{3} s_\gamma c_\gamma g_3^2 \frac{1}{m_b^2} y_e^{i1i2} y_u^{i3i4} LF_{1,1,0}[m_3, m_q^{i3}] + \frac{2}{3} s_\gamma c_\gamma g_3^2 \frac{1}{m_b^2} y_e^{i1i2} y_u^{i3i4} LF_{2,1,-1}[m_3, m_q^{i3}] - \\ & \frac{4}{3} s_\gamma c_\gamma g_3^2 \frac{1}{m_b^2} y_e^{i1i2} y_u^{i3i4} LF_{1,1,0}[m_3, m_u^{i4}] + \frac{2}{3} s_\gamma c_\gamma g_3^2 \frac{1}{m_b^2} y_e^{i1i2} y_u^{i3i4} LF_{2,1,-1}[m_3, m_u^{i4}] - \\ & 3 \frac{1}{m_b^4} y_e^{i1i2} y_u^{i3i4} \left(s_\gamma \tilde{\mu} c_\gamma \overline{y_d^{pr}} \left(-3 s_\gamma c_\gamma a_d^{pr} + \tilde{\mu} y_d^{pr} \left(-2 c_\gamma^2 + s_\gamma^2 \right) \right) + \right. \\ & \quad \left. \overline{a_d^{pr}} \left(s_\gamma c_\gamma a_d^{pr} \left(c_\gamma^2 - 2 s_\gamma^2 \right) + \tilde{\mu} y_d^{pr} \left(c_\gamma^4 - s_\gamma^2 c_\gamma^2 + s_\gamma^4 \right) \right) \right) \\ & \quad LF_{1,1,0}[m_d^r, m_q^p] - \frac{1}{2} s_\gamma c_\gamma \frac{1}{m_b^2} \overline{y_d^{pr}} y_d^{i3r} y_e^{i1i2} y_u^{pi4} LF_{1,1,0}[m_d^r, \tilde{\mu}] - \\ & \frac{1}{m_b^4} y_e^{i1i2} y_u^{i3i4} \left(s_\gamma \tilde{\mu} c_\gamma \overline{y_e^{pr}} \left(-3 s_\gamma c_\gamma a_e^{pr} + \tilde{\mu} y_e^{pr} \left(-2 c_\gamma^2 + s_\gamma^2 \right) \right) + \right. \\ & \quad \left. \overline{a_e^{pr}} \left(s_\gamma c_\gamma a_e^{pr} \left(c_\gamma^2 - 2 s_\gamma^2 \right) + \tilde{\mu} y_e^{pr} \left(c_\gamma^4 - s_\gamma^2 c_\gamma^2 + s_\gamma^4 \right) \right) \right) \\ & \quad LF_{1,1,0}[m_e^r, m_l^p] - \frac{1}{2} s_\gamma c_\gamma \frac{1}{m_b^2} \overline{y_e^{pr}} y_e^{pi2} y_e^{i1r} y_u^{i3i4} LF_{1,1,0}[m_e^r, \tilde{\mu}] + \\ & \frac{1}{m_b^2} y_e^{i1i2} y_u^{i3i4} \left(s_\gamma \tilde{\mu} c_\gamma \overline{y_e^{pr}} \left(-2 s_\gamma c_\gamma a_e^{pr} + \tilde{\mu} y_e^{pr} \left(-c_\gamma^2 + s_\gamma^2 \right) \right) + \right. \\ & \quad \left. \overline{a_e^{pr}} \left(s_\gamma c_\gamma a_e^{pr} \left(c_\gamma^2 - s_\gamma^2 \right) + \tilde{\mu} y_e^{pr} \left(c_\gamma^4 + s_\gamma^4 \right) \right) \right) LF_{2,1,0}[m_l^p, m_e^r] + \\ & \frac{1}{m_b^2} y_e^{i1i2} y_u^{i3i4} \left(s_\gamma \tilde{\mu} c_\gamma \overline{y_e^{pr}} \left(2 s_\gamma c_\gamma a_e^{pr} + \tilde{\mu} y_e^{pr} \left(c_\gamma^2 - s_\gamma^2 \right) \right) + \right. \\ & \quad \left. \overline{a_e^{pr}} \left(s_\gamma c_\gamma$$